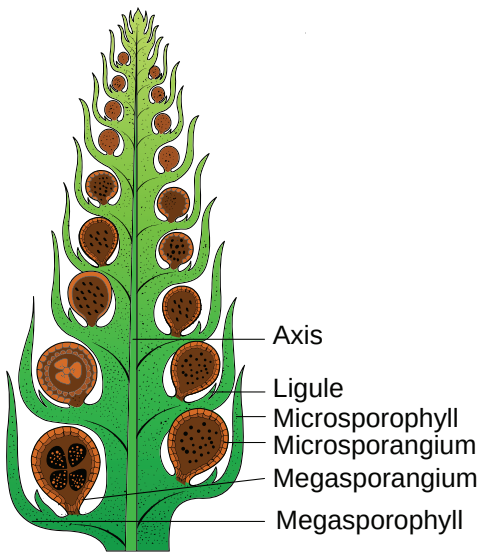
called **trabeculae** is the characteristic feature of *Selaginella*. The casparian strips are found on the lateral walls. The rapid stretching of the innermost cortical cells in comparison with stele results in air spaces and stele appears to be suspended in air space with the help of trabeculae. The stele is a protostele and exarch. A variation in number of steles is found. It may be monostelic (*S. spinulosa*); distelic (*S. kraussiana*)or polystelic (*S. laevigata*). The xylem is monarch(*S. kraussiana*) or diarch (*S. oregana*). Tracheids are present but vessels are also noticed in *S. densa* and *S. rupestris*.

Leaf

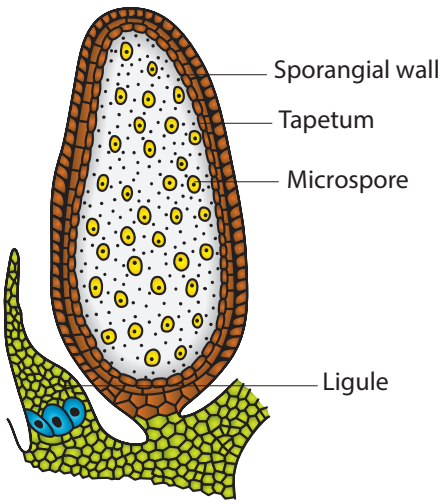
The leaf shows upper and lower epidermis. The epidermal cells have chloroplast. Stomata occur on both surfaces. The mesophyll is made up of loosely arranged thin walled cells with intercellular spaces. There is a median vascular bundle surrounded by a bundle sheath. In vascular bundle xylem is surrounded by phloem.

Reproduction

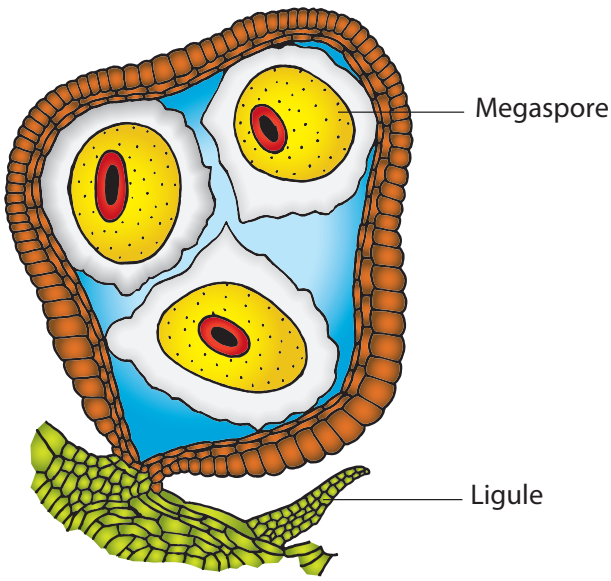
Selaginella shows both vegetative and asexual modes of reproduction.



a) L.S. of cone



b) A microsporangium enlarged



c) A megasporangium enlarged

Figure 2.29: Reproduction in *Selaginella*

Vegetative reproduction

Selaginella reproduces vegetatively by fragmentation, bulbil formation, tuber formation and resting buds.

Sexual reproduction

During sexual reproduction spores are produced (Figure 2.29). *Selaginella* is heterosporous and produces two types of spores namely microspores in